

Geometric and parametric control of the Diode-effect in an asymmetric superconducting sample

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This study investigates the Superconducting Diode Effect (SDE) in two-dimensional superconducting films with asymmetric comb-like edge geometries using the generalized time-dependent Ginzburg-Landau (gTDGL) equations. We comprehensively analyze the rectification efficiency as a function of geometric topology, external magnetic field intensity, and the Ginzburg-Landau parameter. Our findings reveal that non-reciprocity is not achieved through a linear scaling of edge defects, but via a strict resonance between the electromagnetic penetration depth, the field-induced vortex lattice separation, and the device's asymmetric topology. We identify two distinct operational regimes: at low magnetic fields, intermediate defect densities optimize directional vortex trapping and current crowding, yielding robust asymmetries; at high fields, geometries with larger internal free volume better accommodate dense vortex lattices, preventing premature surface barrier saturation and retaining high efficiencies. Furthermore, high Ginzburg-Landau parameters suppress the effective spatial asymmetry by altering the screening currents. Ultimately, we demonstrate that optimal geometric design alone can generate substantial rectification at minimal magnetic fields without requiring remanent magnetization, offering critical insights for the development of ultra-low-power superconducting electronics.

Keywords: Superconducting diode effect; Vortex dynamics; Time-dependent Ginzburg-Landau; Mesoscopic; superconductor.

I. INTRODUCTION

In recent years, the field of superconducting electronics has experienced a significant resurgence driven by the discover and characterization of the Superconducting Diode Effect (SDE). This phenomenon, analogous to rectification in semiconductors but with potentially zero energy dissipation, has opened new avenues for the development of ultra-low-power logic and memory circuits [1], [2], [3]. Fundamentally, the SDE manifests as a non-reciprocity in the critical current (I_c), where the magnitude of the current required to transition from the superconducting to the normal state differs depending on the polarity of the charge flow ($I_c^+ \neq I_c^-$). For this non-reciprocity to emerge, breaking both spatial inversion symmetry (IS) and time-reversal symmetry (TRS) is a necessary condition [4], [5]. Recent research has focused intensely on discerning whether the origin of this symmetry breaking (and thus the SDE) stems from intrinsic characteristics of the band structure and spin-orbit coupling, of from extrinsic factors related to device geometry and vortex dynamics [6], [7]. While intrinsic mechanisms such as the finite momentum of Cooper pairs are explored in systems like topological semimetals or artificial super-

lattices [6], [1], other studies have demonstrated that geometric design and deformation can effectively control the diode response [8], [7]. To extrapolate from these investigations and relate observable macroscopic results to microscopic and mesoscopic causes, several authors have adopted the theoretical framework of the Ginzburg-Landau (GL) equations and their time-dependent extensions (TDGL and gTDGL) [9], [10]. This conceptual approach enables the modeling of vortex matter dynamics and screening currents in complex geometries, offering a deep phenomenological understanding that is often difficult to captured solely through microscopic models [11], [12]. For instance, the use of generalized TDGL equations has been crucial to describe how the asymmetric nucleation of vortices and antivortices, in the presence of defects or inhomogeneous fields, leads to a non-reciprocal resistance [13], [14]. In the quest to maximize the rectification efficiency η , multiple works have explored diverse topologies, including asymmetric antidot lattices [15], notches or edge defects [16], [17], and conformal arrays [15][18]. While antidots allow for distributed flux control, recent results indicate that edge defects tend to be more efficient by exploiting the local suppression of the Bean-Livingston barrier and the current crowding effect [17]. Even more notable are conformal arrays, where the spatial gradient in pinning density induces a "funnel" effect, achieving rectification signals orders of magnitude larger than those of conventional flux diodes [18]. Crucially, recent literature has revealed through simulations the for-

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mation of "flux pockets" between edge indentations [17]. This mechanism acts as an asymmetric trap where vortices remain confined in one direction of the current but flow freely in the opposite, a finding that underpins the geometry and approach adopted in this research. One of the most relevant implications of these findings is the pursuit of maximum rectification efficiency, striving toward the ideal of a superconducting diode that maintains zero resistance in one direction while presenting high impedance in the other [19], [11]. Recently, attention has shifted not only toward instantaneous efficiency but also toward the device's memory functionality. Pioneering studies have demonstrated diodes with programmable memory capabilities via magnetic flux trapping, allowing for switchable non-reciprocity [20]. Furthermore, high-stability and high-efficiency diode effects have been reported in high-temperature superconductors, holding promise for state retention in practical applications [21]. In this context, the present work employs the framework of the generalized time-dependent Ginzburg-Landau (gTDGL) equations to theoretically investigate the diode effect. Broadly, this study focuses on determining the efficiency of the diode effect and evaluating memory capacity as a function of specific device geometry. Guided by the previously cited literature, we capitalize on the flux pocket mechanism to optimize design parameters for future applications in fluxonics and superconducting computing.

A. Sample Geometry and Experimental Design

The simulation configuration models a two-dimensional Type-II superconducting film structured with an asymmetric comb-shaped geometry. The device architecture is defined within a rectangular domain Ω bounded by a contour $\partial\Omega$, with a total length L_x of 20 μm , a block height L_y of 5.0 μm , and a thickness L_z of 0.1 μm . The geometry features a series of indentations (teeth) along the lower edge, characterized by a defect depth h_d of 3.5 μm (see Fig. 1). The density and profile of the teeth were modeled using a superposition of Gaussian functions. The lower edge profile $y(x)$ is defined as:

$$y(x) = -h_d \sum_{i=1}^N \exp\left(-\frac{(x-x_i)^2}{2\delta^2}\right) \quad (1)$$

where h_d represents the maximum depth of the teeth, x_i denotes the central position of the i -th tooth, and δ is the geometric smoothness parameter. To maintain geometric self-similarity across the different configurations, δ is defined as a fraction of the inter-tooth separation Δx , specifically $\delta = 0.20\Delta x$. This formulation allows for the discrete variation of the number of teeth $N \in \{4, 8, 12, 16\}$, which consequently alters both the separation Δx and the steepness of the potential landscape, as illustrated in Fig. 2. For the transport char-

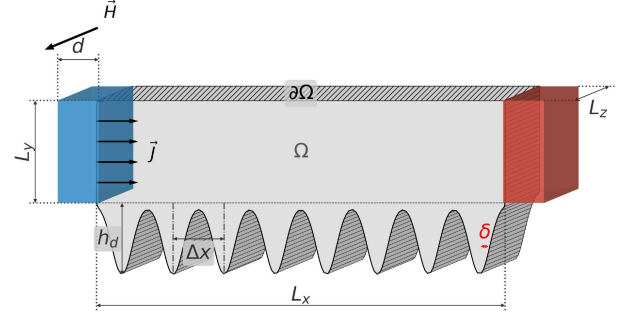


FIG. 1. 3D schematic representation of the asymmetric comb-shaped superconducting device geometry. The superconducting domain (Ω , gray) is bounded by its boundary ($\partial\Omega$). The source (blue) and drain (red) terminals are indicated. An applied external magnetic field, $\vec{H}$, and a uniformly applied current density, $\vec{J}$, are denoted by vectors. The magnetic field $\vec{H}$ is applied at a distance d from the source terminal. The main dimensions are the total length L_x (20 μm), the strip height L_y (5.0 μm), and the thickness L_z (0.1 μm). The lower profile features Gaussian indentations (teeth) with a maximum depth h_d (3.5 μm), inter-tooth separation Δx , and a geometric parameter δ . The figure illustrates a configuration with a specific number of teeth ($N = 8$).

acterization, the geometric configurations are subjected to an external direct current (DC) density $\mathbf{J}$. This current is injected through two metallic contacts of length d equal to 2.0 μm (source and drain terminals) and is applied along the x -axis, under the influence of a perpendicular magnetic field $\mathbf{H} = H\hat{z}$. The analysis of the current-voltage ($I - V$) characteristic curves allows for the determination of the critical currents (J_c^+ and J_c^-) and the calculation of the Superconducting Diode Effect (SDE) rectification efficiency. The central objective is to elucidate the correlation between the density of geometric defects (number of teeth N) and the rectification efficiency, along with its memory capacity through the analysis of hysteresis curves. Finally, we analyze the interplay between vortex dynamics and the energy landscape modulated by this variable geometry.

II. THEORETICAL FORMALISM

In the present work, numerical simulations were carried out using the open-source package `pyTDGL` [22], which solves the generalized time-dependent Ginzburg-Landau (gTDGL) equations in two dimensions. Our study system, illustrated in Figs. 1 and 2, consists of three two-dimensional mesoscopic superconducting samples occupying a domain Ω . These samples exhibit a dented geometry characterized by a number of teeth N_i (with $N \in \{4, 8, 12, 16\}$), modeled via a Gaussian function (Eq. 1). These geometries induce a reflection symmetry breaking with respect to the y -axis (σ_y). The samples are subjected to a uniform external magnetic field applied in

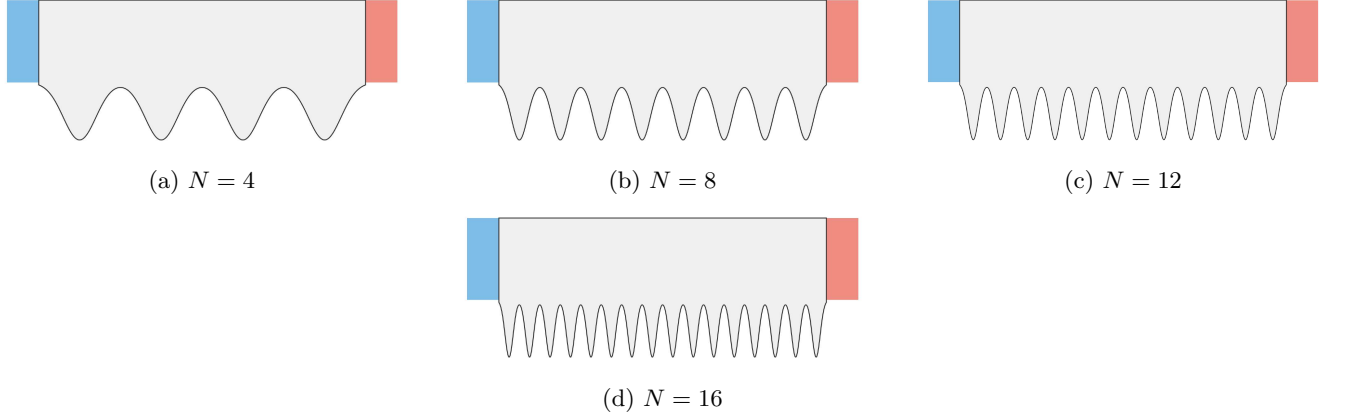


FIG. 2. Evolution of the superconducting device geometry as a function of the number of Gaussian teeth (N). Four configurations are shown (from left to right): (a) **Device 1** ($N = 4$), with profile $y(x) = -3.5 \sum_{i=1}^4 \exp \left[-\frac{(x-x_i)^2}{2(1.00)^2} \right]$; (b) **Device 2** ($N = 8$), with profile $y(x) = -3.5 \sum_{i=1}^8 \exp \left[-\frac{(x-x_i)^2}{2(0.50)^2} \right]$; (c) **Device 3** ($N = 12$), with profile $y(x) = -3.5 \sum_{i=1}^{12} \exp \left[-\frac{(x-x_i)^2}{2(0.33)^2} \right]$; and (d) **Device 4** ($N = 16$), with profile $y(x) = -3.5 \sum_{i=1}^{16} \exp \left[-\frac{(x-x_i)^2}{2(0.25)^2} \right]$. The lower edge profile is generated via the superposition of Gaussian functions of constant depth $h_d = 3.5 \mu\text{m}$, where the smoothness parameter δ decreases proportionally as the number of teeth increases.

the $\hat{z}$ direction, with variable magnitude $\mathbf{H} = H\hat{z}$, and to an external transport current applied in the $\hat{x}$ direction with variable density $\mathbf{J} = J\hat{x}$. Current injection is performed through metallic contacts located on the lateral edges (left and right) of the sample. The boundary between the superconducting region and the vacuum or contacts is denoted by $\partial\Omega$. The temporal evolution of the superconducting state is described by the complex order parameter $\psi(\mathbf{r}, t)$ and the electrostatic potential $\mu(\mathbf{r}, t)$. The model is based on the gTDGL equations obtained from the minimization of the Gibbs free energy. In dimensionless units, the equation governing the dynamics of ψ is:

$$\frac{u}{\sqrt{1 + \gamma^2 |\psi|^2}} \left[\frac{\partial \psi}{\partial t} + i\mu\psi + \frac{\gamma^2}{2} \frac{\partial |\psi|^2}{\partial t} \psi \right] = (\nabla - i\mathbf{A})^2 \psi + \psi(\epsilon(\mathbf{r}) - |\psi|^2), \quad (2)$$

where $u = \pi^4/14\zeta(3) \approx 5.79$ represents the ratio of relaxation times for the amplitude and phase of the order parameter in the dirty limit ($\zeta(x)$ is the Riemann zeta function), and γ parameterizes the strength of inelastic electron-phonon scattering. Throughout this work, we assume $\gamma = 10$. The scalar function $\epsilon(\mathbf{r}) = T_c(\mathbf{r})/T - 1 \in [-1, 1]$ adjusts the local critical temperature of the film. The value $\epsilon(\mathbf{r}) = 1$ simulates an ideal superconducting region, while $\epsilon(\mathbf{r}) < 1$ locally suppresses superconductivity. Extended regions with $\epsilon(\mathbf{r}) < 0$ are used to model defects or metallic inclusions. The total current density $\mathbf{J}$ is defined as the sum of the supercurrent density $\mathbf{J}_s$ and the normal current density $\mathbf{J}_n$:

$$\mathbf{J} = \mathbf{J}_s + \mathbf{J}_n = \text{Im}[\psi^*(\nabla - i\mathbf{A})\psi] - \nabla\mu. \quad (3)$$

Applying the current continuity condition $\nabla \cdot \mathbf{J} = 0$ to Eq. (3), we obtain:

$$\nabla \cdot \mathbf{J} = \nabla \cdot (\mathbf{J}_s + \mathbf{J}_n) = \nabla \cdot \text{Im}[\psi^*(\nabla - i\mathbf{A})\psi] - \nabla^2 \mu = 0.$$

This leads to the Poisson equation describing the electric potential distribution $\mu(\mathbf{r}, t)$:

$$\nabla^2 \mu = \nabla \cdot \text{Im}[\psi^*(\nabla - i\mathbf{A})\psi] = \nabla \cdot \mathbf{J}_s. \quad (4)$$

For thin films (thickness $d \ll \lambda$), the relevant quantities are the sheet current densities (integrated over the thickness), denoted as $\mathbf{K} = \mathbf{K}_s + \mathbf{K}_n = d(\mathbf{J}_s + \mathbf{J}_n)$.

A. Boundary Conditions and Units

Since Eq. (4) constitutes a pure Neumann problem for μ , the solution is determined up to an additive constant. We fix the reference by imposing that the average potential at the output contact is zero, $\langle \mu \rangle_{\partial\Omega_{out}} = 0$. The instantaneous voltage $V(t)$ is calculated as the potential difference between the electrodes:

$$V(t) = \langle \mu \rangle_{\partial\Omega_{in}} - \langle \mu \rangle_{\partial\Omega_{out}}, \quad (5)$$

where $\langle \mu \rangle_S = \frac{1}{|S|} \int_S \mu \, dS$ denotes the spatial average over the surface S of the contact. The DC voltage is obtained by time-averaging $V(t)$ over a stationary time window. The boundary conditions for the scalar potential are $\hat{n} \cdot \nabla \mu = \mp J_{ext}$ at the input and output metallic contacts, and $\hat{n} \cdot \nabla \mu = 0$ at the superconductor-vacuum interfaces, with $\hat{n}$ being the normal vector to the surface. For the order parameter, we employ mixed

boundary conditions: at the metallic contacts, we impose $\psi = 0$ (simulating normal metal interfaces), while at the rest of the superconductor-vacuum boundary, we apply the Neumann condition $\hat{n} \cdot (\nabla - i\mathbf{A})\psi = 0$. The equations are solved in dimensionless units: distances are scaled by the coherence length ξ ; time by $\tau_0 = \mu_0\sigma\lambda^2$; the magnetic field by $H_{c2} = \Phi_0/2\pi\xi^2$; and the vector potential by $A_0 = \Phi_0/2\pi\xi$. The current density is normalized to $J_0 = 4\xi H_{c2}/\mu_0\lambda^2$, the sheet current to $K_0 = J_0d = 4\xi B_{c2}/\mu_0\Lambda$ (where $\Lambda = \lambda^2/d$ is the effective Pearl penetration depth), and the voltage to $V_0 = 4\xi^2 H_{c2}/\mu_0\sigma\lambda^2$. Here, $\Phi_0 = h/2e$ is the magnetic flux quantum. For the simulations, we fix $\xi = 0.5\mu\text{m}$. We consider Type-II superconductors characterized by the Ginzburg-Landau parameter $\kappa = \lambda/\xi$. The values used for κ and H_{c2} are consistent with previous studies on mesoscopic systems.

B. Numerical Implementation

The numerical solution of the coupled gTDGL equations is implemented using the finite volume method on an unstructured Delaunay triangular mesh. The dual Voronoi mesh provides a natural discretization that guarantees flux conservation, where spatial differential operators are approximated by evaluating fluxes across the edges of the Voronoi cells [23], [24]. At each time step, the solution scheme follows an adaptive Euler method structured as follows:

1. **Order parameter update (ψ):** $\psi(\mathbf{r}, t^{n+1})$ is calculated. To ensure numerical stability, especially when $|\psi|^2 \rightarrow 0$, pyTDGL solves an implicit quadratic equation derived from the temporal discretization, selecting the corresponding physical root to update the state.
2. **Current calculation:** With the new value of ψ , the supercurrent density $\mathbf{J}_s$ and its divergence are recalculated.
3. **Electric potential resolution (μ):** The discrete Poisson equation (Eq. 4) is solved to obtain $\mu(\mathbf{r}, t^{n+1})$. Unlike simple iterative methods, the solver uses a Sparse LU factorization to solve the linear system efficiently and robustly.
4. **Adaptive time-step control:** A dynamic algorithm is employed for Δt . The time step is adjusted based on the rate of change of $|\psi|^2$. If an update results in a non-physical state (e.g., negative discriminant in the update equation), the algorithm iteratively reduces Δt until stability is recovered.

The implementation of pyTDGL is grounded in the theoretical and numerical developments established in seminal works such as those by Kramer and Watts-Tobin [25], [26], and has been validated in recent studies of superconducting nanodevices [27], [22].

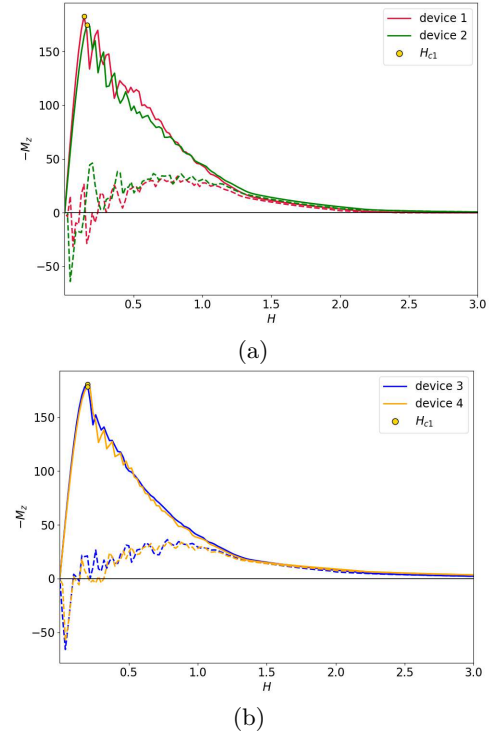


FIG. 3. Magnetization curve ($-M_z$) as a function of the applied external magnetic field H for a Ginzburg-Landau parameter $\kappa = 2$. (a) Hysteresis loops for Device 1 and Device 2. (b) Hysteresis loops for Device 3 and Device 4. The first critical field H_{c1} and the maximum diamagnetic response ($-M_{\text{peak}}$) obtained are: $H_{c1} = 0.14$, $-M_{\text{peak}} = 183.01$ for Device 1; $H_{c1} = 0.18$, $-M_{\text{peak}} = 176.18$ for Device 2; $H_{c1} = 0.20$, $-M_{\text{peak}} = 180.20$ for Device 3; and $H_{c1} = 0.20$, $-M_{\text{peak}} = 178.20$ for Device 4.

III. MEISSNER EFFECT

The analysis of the Meissner effect allows us to determine the magnetic response of the material at low temperatures under the influence of varying external magnetic fields. This facilitates the observation of the three fundamental thermodynamic states: the superconducting state, the mixed (or transition) state, and the normal state. In this context, the first critical magnetic field, H_{c1} , is defined as the point at which the magnetization of the material reaches its maximum magnitude, denoted by $-M_{\text{peak}}$. This point marks the loss of perfect diamagnetism due to the initial penetration of vortices into the sample. The evolution of the material's response to the external magnetic field H is illustrated in Figs. 3, 4, and 5; to evaluate the memory capacity and vortex trapping, the devices are initially subjected to an increasing magnetic field from a zero-field cooled (ZFC) state (solid lines) until the magnetization approaches zero, indicating the onset of the normal state. Subsequently, the applied field is progressively reduced back to zero (dashed lines). Table III summarizes the calculated values of

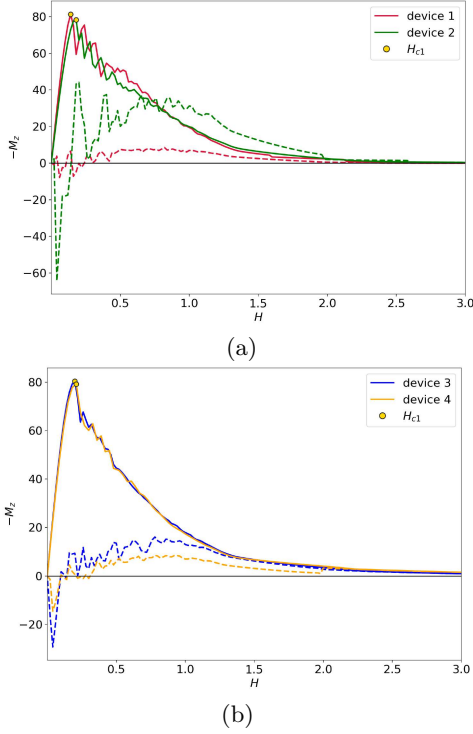


FIG. 4. Magnetization curve ($-M_z$) as a function of the applied external magnetic field H for a Ginzburg-Landau parameter $\kappa = 3$. (a) Hysteresis loops for Device 1 and Device 2. (b) Hysteresis loops for Device 3 and Device 4. The first critical field H_{c1} and the maximum diamagnetic response ($-M_{\text{peak}}$) obtained are: $H_{c1} = 0.14$, $-M_{\text{peak}} = 81.34$ for Device 1; $H_{c1} = 0.18$, $-M_{\text{peak}} = 78.30$ for Device 2; $H_{c1} = 0.20$, $-M_{\text{peak}} = 80.37$ for Device 3; and $H_{c1} = 0.21$, $-M_{\text{peak}} = 79.20$ for Device 4.

the critical fields H_{c1} , as well as the magnetization peak, for the three superconducting devices. Analyzing these data, it is clearly observed that the first critical field, H_{c1} , shows no dependence on the Ginzburg-Landau parameter, κ . In contrast, the maximum magnetization ($-M_{\text{peak}}$) exhibits a strong dependence on this parameter, explicitly showing that as κ increases, the magnitude of the magnetization decreases significantly. In fact, the magnitude of the change in the maximum magnetization values is influenced to a much greater degree by κ than by the variation in the device geometry itself. This behavior provides a crucial first impression of the geometric sensitivity of the system, indicating that, although the topology affects the local dynamics, the intrinsic properties of the material dominate the maximum diamagnetic shielding capacity. Furthermore, the closing of the hysteresis loops at zero applied field indicates the absence of significant remanent magnetization, implying that these specific geometric configurations do not exhibit a macroscopic memory effect at zero field.

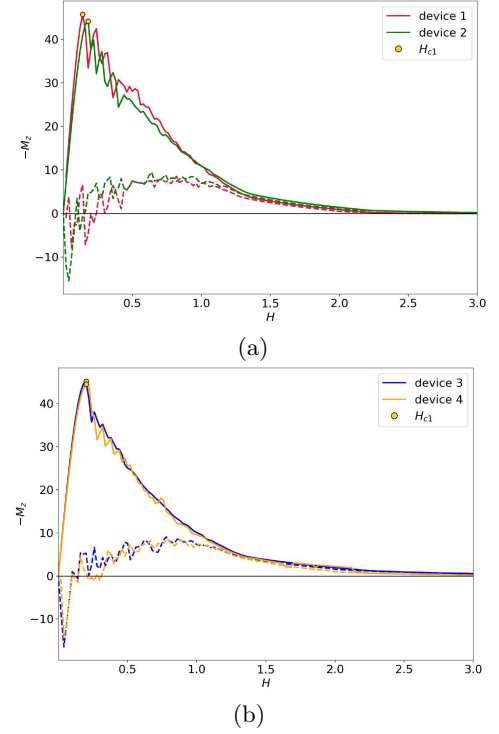


FIG. 5. Magnetization curve ($-M_z$) as a function of the applied external magnetic field H for a Ginzburg-Landau parameter $\kappa = 4$. (a) Hysteresis loops for Device 1 and Device 2. (b) Hysteresis loops for Device 3 and Device 4. The first critical field H_{c1} and the maximum diamagnetic response ($-M_{\text{peak}}$) obtained are: $H_{c1} = 0.14$, $-M_{\text{peak}} = 45.75$ for Device 1; $H_{c1} = 0.18$, $-M_{\text{peak}} = 44.16$ for Device 2; $H_{c1} = 0.20$, $-M_{\text{peak}} = 45.21$ for Device 3; and $H_{c1} = 0.20$, $-M_{\text{peak}} = 44.55$ for Device 4.

IV. SUPERCONDUCTING DIODE EFFECT (SDE)

In this section, the phenomenology of the Superconducting Diode Effect (SDE) is comprehensively analyzed along with its simultaneous dependence on three fundamental variables: the geometric topology of the device (responsible for the magnitude of the spatial inversion symmetry breaking), the intensity of the applied external magnetic field (inductor of the time-reversal symmetry breaking), and the modulation of the Ginzburg-Landau parameter (κ). The implemented computational methodology evaluates four device topologies with comb-like geometry (differentiated by their number and distribution of indentations or "teeth"). For each design, the intrinsic parameter of the material is varied, adopting the values of $\kappa \in \{2, 3, 4\}$. Subsequently, each configuration is subjected to an external transport current under the influence of a static perpendicular magnetic field, varying in the interval $H \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ mT. This parametric sweep generates 60 independent measurements of the SDE efficiency, summarized in Table I,

Critical field (H_{c1}) and maximum magnetization peak (M_{peak}) for different device configurations and Ginzburg-Landau parameters (κ).

κ	Device	$H_{c1}[\text{mT}]$	$M_{\text{peak}}[\mu\text{A}/\mu\text{m}^3]$
2	1	0.14	183.01
	2	0.18	176.18
	3	0.20	180.20
	4	0.20	178.20
3	1	0.14	81.34
	2	0.18	78.30
	3	0.20	80.37
	4	0.21	79.20
4	1	0.14	45.75
	2	0.18	44.16
	3	0.20	45.21
	4	0.20	44.55

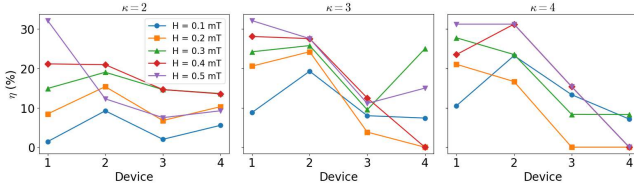


FIG. 6. Global comparison of the SDE efficiency as a function of the geometric topology of the evaluated devices.

II, III, and obtained through Equation 6. It is imperative to highlight that the criterion for determining the critical currents (J_c^+ and J_c^-) was based on the differential analysis of the voltage-current curves (dV/dJ), identifying the point of maximum differential resistance. This methodology guarantees the strict reproducibility of the simulated experimental data by mitigating observational biases. Likewise, Table ?? shows the values of the first critical current for both polarizations, J_{c1}^+ and J_{c1}^- , which is defined as the current at which the mixed state begins and, therefore, the loss of the superconducting state; as well as the second critical current for both polarizations, J_{c2}^+ and J_{c2}^- .

$$\eta = \frac{||J_c^+| - |J_c^-||}{|J_c^+| + |J_c^-|} \times 100 \quad (6)$$

The graphic analysis focuses on the representative configurations that exhibit the best efficiency results, which correspond to Device 1 ($\kappa = 3, H_{c1} = 0.14$, Fig. 7) and Device 2 ($\kappa = 3, H_{c1} = 0.18$, Fig. 8).

A. Parametric Dependence of the SDE Efficiency

As a starting point for the analysis of Figure 6, it is fundamental to consider the direct proportionality relationship between the Ginzburg-Landau parameter κ and the penetration depth λ , defined as $\lambda = \kappa\xi$. Fixing the

coherence length at $\xi = 0.5 \mu\text{m}$, an increase in κ results in a greater spatial extent of the magnetic screening currents. Consequently, for high values of κ (specifically $\kappa = 4$), the penetration depth becomes comparable to or greater than the separation between indentations in the denser geometries. This electromagnetic overlapping causes the system to perceive the lower contour not as a dented array, but effectively as a smooth edge. This explains why the rectification efficiency η abruptly decays to values close to zero for Devices 3 and 4 under the $\kappa = 4$ configuration. Therefore, to narrow the analysis and observe the real impact of the geometric topology without the masking produced by the penetration depth, the following results focus on Devices 1 and 2. When evaluating Device 1, the results reveal a predominantly linear correlation between the efficiency η and the applied magnetic field H , with the notable exception of a local anomaly for the $\kappa = 4$ configuration at $H = 0.4 \text{ mT}$, where this linearity is broken. An outstanding feature of this first topological configuration is its ability to maximize the diode effect in the high magnetic field regime. Specifically, when subjecting the sample to a field $H = 0.5 \text{ mT}$, the device achieves robust efficiencies of around 30%, a behavior that remains remarkably stable and independent of the selected κ value. In stark contrast, Device 2 exhibits opposite behavior regarding magnetic intensity: its efficiency degrades in high-field regimes but is significantly optimized for the low magnetic field regime. In this design, the efficiency thresholds consistently exceed those recorded by Device 1 for fields of $H \in \{0.1, 0.2, 0.3\} \text{ mT}$. Additionally, in this low-field regime, a constructive dependence on the Ginzburg-Landau parameter emerges, where an increase in κ directly favors the magnitude of the asymmetry. A clear example of this behavior is observed in the response of Device 2 to a field of $H = 0.1 \text{ mT}$, where the efficiency experiences a considerable jump as κ increases, recording values of 9.25% for $\kappa = 2$, 19.35% for $\kappa = 3$, and reaching a maximum of 23.53% for $\kappa = 4$. The increase in the parameter κ decreases the global tolerance of the material by increasing the penetration of the Meissner currents (J_M). However, this collapse in transport capacity is drastically asymmetric due to the geometry, a topic that will be explained in the following section, where we will understand why κ amplifies the efficiency at low fields in Device 2.

V. ORDER PARAMETER AND ENERGY LANDSCAPE

In the previous section, the parametric dependencies of the system were established, but it is imperative to understand the underlying physical and electromagnetic mechanisms that give rise to the superconducting diode effect. To this end, we topologically analyze the order parameter and the spatial distribution of the currents. Of particular interest is the behavior of the devices exposed

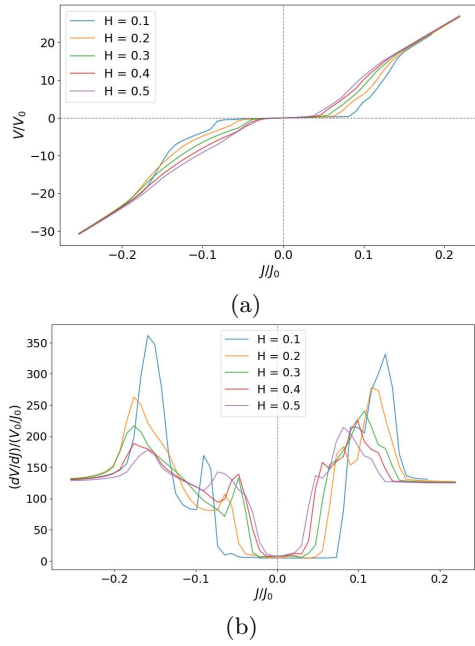


FIG. 7. Transport characteristics for Device 1 with $\kappa = 3$ ($H_{c1} = 0.14$). (a) Voltage-current ($V - J$) relationship parametric in the magnetic field H . (b) Differential resistance (dV/dJ) used for the precise extraction of the critical currents J_c^+ and J_c^- .

to a low external magnetic field, specifically $H = 0.1$ mT. As detailed in Table ??, this value is below the first critical field (H_{c1}). In this regime, Device 2 demonstrates that an increase in the parameter κ translates into a significant increase in efficiency. This behavior establishes *a priori* that it is possible to generate a robust diode effect relying purely on geometry, without the need to subject the sample to magnetic fields near or above the critical field. To elucidate this phenomenon, the vortex distribution in Devices 1 and 2 (fixing $\kappa = 3$) is analyzed by varying the transport current within the mixed state regime for $H \in \{0.1, 0.3, 0.5\}$ mT. The fundamental question lies in understanding why such marked non-reciprocity emerges if the magnitude of the injected current is symmetric. The answer lies in the electromagnetic interaction between the geometric asymmetry of the device and the superposition of the currents. Under an external magnetic field H , the sample induces a Meissner screening current (J_M). Upon injecting an external transport current (J), the total current distribution of the system is dictated by their vectorial superposition ($J_{total} = J_M \pm J$). Due to the geometry, this distribution is highly heterogeneous: the smooth upper edge maintains a quasi-homogeneous current profile, while the dented lower edge experiences severe fluctuations. As evidenced in the current and order parameter maps (Fig. 9), the dynamics of the system exhibit a radical bifurcation depending on the polarity of J . The asymmetry in the Lorentz force ($\vec{F}_L = \vec{J} \times \vec{B}$) dictates the direction of

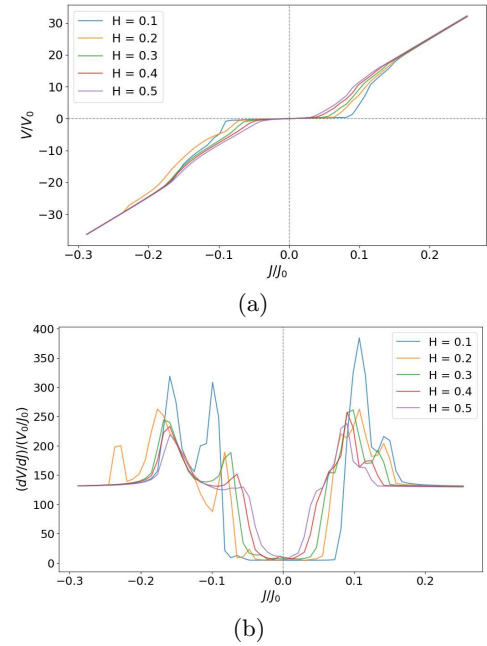


FIG. 8. Transport characteristics for Device 2 with $\kappa = 3$ ($H_{c1} = 0.18$), the configuration that maximizes asymmetry at low magnetic fields. (a) Voltage-current ($V - J$) relationship. (b) Differential resistance (dV/dJ) used for the precise extraction of the critical currents J_c^+ and J_c^- .

vortex nucleation and movement, generating two clearly differentiated transport regimes:

1. **Positive Polarity ($J > 0$, vortices toward $-\hat{y}$):** The Lorentz force pushes the vortices from the smooth upper edge toward the dented lower edge. Lacking preferential nucleation sites at the top, vortices enter in a disordered manner. As they transit through the *bulk* toward the dented edge, the narrowing of the cross-section induces a strong *current crowding* phenomenon. In this regime, the *vortex paths* take longer to structure and lack strong pinning, resulting in a premature and abrupt collapse into the normal state.
2. **Negative Polarity ($J < 0$, vortices toward $+\hat{y}$):** The Lorentz force reverses its direction, forcing the nucleation of vortices at the lower edge. The indentations act as electromagnetic stress concentrators, becoming preferred entry points. This geometry facilitates the rapid formation of highly structured and parallel *vortex paths*. The strong trapping provided by the geometry significantly delays dissipation, allowing the superconducting state to withstand much larger current magnitudes before collapsing.

An explicit contrast of this phenomenology is observed when comparing the order parameter maps for $I_+ = 34 \mu\text{A}$ and $I_- = -31.33 \mu\text{A}$ in Figure 11. Furthermore, it should be noted that the higher number of teeth in

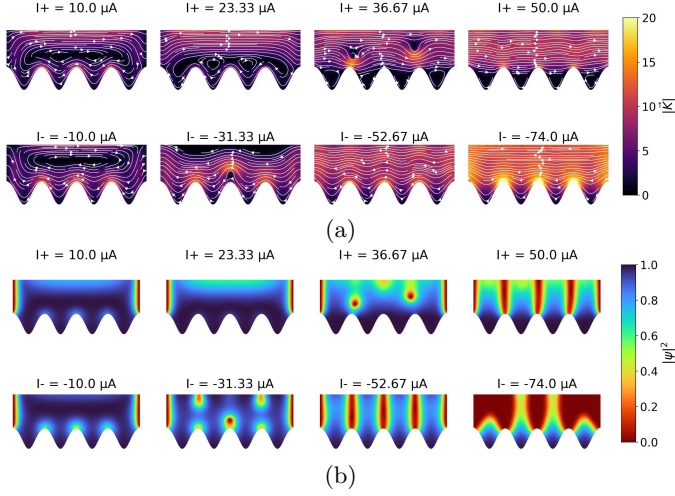


FIG. 9. Spatial distribution of (a) the supercurrent density and (b) the superconducting order parameter for Device 1 ($\kappa = 3$) exposed to a static magnetic field of $H = 0.1$ mT. The matrices contrast the evolution of the mixed state, evidencing the asymmetry in vortex nucleation and the *current crowding* phenomenon upon reversing the polarity of the transport current.

Device 2 induces an earlier organization of the vortex river structures (vortex paths). This early structuring is clearly evident when comparing the order parameter maps of Devices 1 and 2 with field $H = 0.5$ (Figs. 13 and 14) at currents such as $I_- = -10 \mu\text{A}$ and $I_+ = 19.33 \mu\text{A}$. Finally, this coupled dynamics also explains the localized zones of stationary current density within the tooth cavities (observable in the positive polarity). These circular currents do not require the presence of a vortex core; rather, they are the product of the screening currents confined by the topology of the device, altering the local energy landscape.

VI. CONCLUSIONS

In this work, the rectification efficiency of the Superconducting Diode Effect (SDE) in thin films with dented edge geometries (comb-like) was investigated using the generalized time-dependent Ginzburg-Landau (gTDGL) equations. The results demonstrate that the magnitude and stability of the non-reciprocity do not respond to a linear scaling of the number of defects, but rather to a strict resonance between the electromagnetic penetration depth (λ), the vortex lattice separation induced by the magnetic field (H), and the asymmetric topology of the device. It was determined that the Ginzburg-Landau parameter (κ) acts as a critical modulator of efficiency by altering the spatial distribution of the screening currents. For high values of κ , where λ becomes comparable to the physical spacing between the indentations, the Meiss-

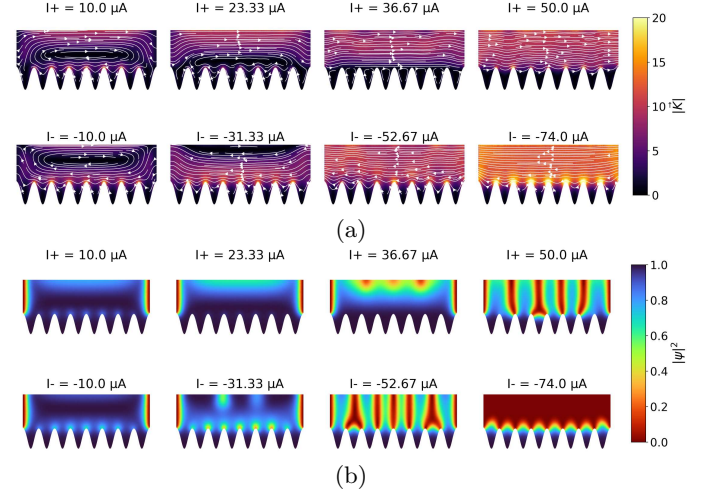


FIG. 10. Spatial distribution of (a) the supercurrent density and (b) the superconducting order parameter for Device 2 ($\kappa = 3$) under a field of $H = 0.1$ mT. The visualization demonstrates how the topology of the dented edge facilitates strong directional trapping and the structured formation of *vortex paths*, mechanisms responsible for maximizing asymmetry at low magnetic fields.

ner currents are unable to contour the individual defects. Consequently, the system perceives a quasi-smooth lower edge, which destroys the effective spatial asymmetry and causes the rectification efficiency to collapse, as evidenced in the higher density geometries (Devices 3 and 4). Furthermore, two mutually exclusive operating regimes dictated by the magnetic field were identified. In the low-field regime ($H \leq 0.3$ mT), the efficiency is governed almost exclusively by the geometric density. Configurations with intermediate density (Device 2) achieve optimal trapping, facilitating the rapid formation of stable *vortex paths* under a negative polarity ($J < 0$, with vortex motion towards $+\hat{y}$). In contrast, for the opposite polarity ($J > 0$, with vortex motion towards $-\hat{y}$), the geometric narrowing induces severe *current crowding* at the tips, causing a premature transition to the normal state. On the other hand, in the high magnetic field regime ($H = 0.5$ mT), repulsive vortex-vortex interaction dominates the dynamics. Dense geometries suffer a premature saturation of the surface barrier, whereas geometries with a larger free volume or internal *bulk* (Device 1) manage to accommodate the dense vortex lattice without sacrificing edge asymmetry, retaining high rectification efficiencies of up to 32%. Finally, the order parameter analysis confirms that the SDE in these mesoscopic systems does not require fields close to the thermodynamic limit H_{c1} . The geometry alone, when coupled with the asymmetry of the Lorentz force, is capable of locally redirecting the vortex flow and generating asymmetries in critical currents greater than 20% at minimal fields (0.1 mT). This provides a fundamental design criterion for future ultra-

low-power superconducting electronics architectures, optimizing the topology to maximize non-reciprocity without requiring remanent magnetization.

APPENDIX A. VISUALIZATION OF THE CURRENT AND ORDER PARAMETER FOR DEVICES 1 AND 2 WITH $\kappa = 3$ AT EXTERNAL MAGNETIC FIELDS OF 0.3 mT AND 0.5 mT

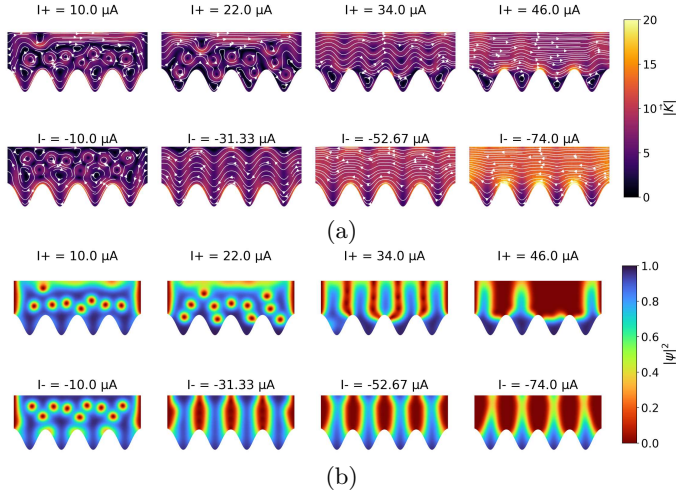


FIG. 11. Spatial distribution of (a) the supercurrent density and (b) the superconducting order parameter for Device 1 ($\kappa = 3$) exposed to a static magnetic field of $H = 0.3$ mT.

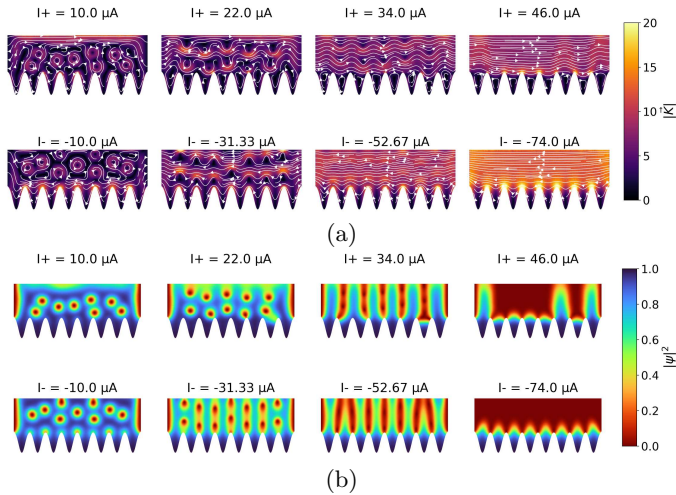


FIG. 12. Spatial distribution of (a) the supercurrent density and (b) the superconducting order parameter for Device 2 ($\kappa = 3$) exposed to a static magnetic field of $H = 0.3$ mT.

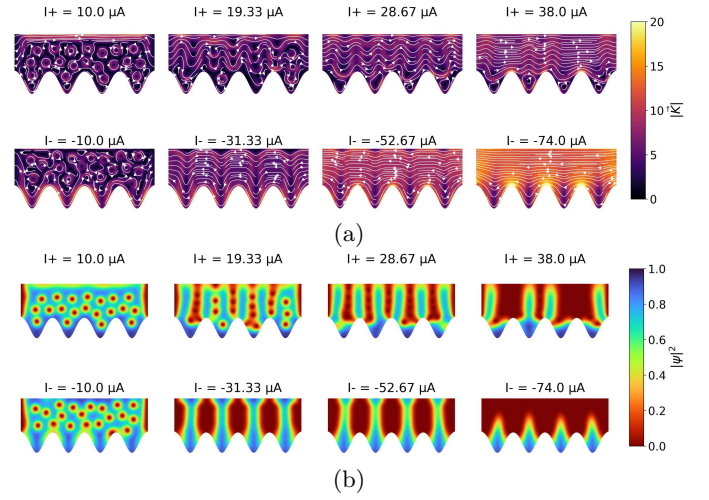


FIG. 13. Spatial distribution of (a) the supercurrent density and (b) the superconducting order parameter for Device 1 ($\kappa = 3$) exposed to a static magnetic field of $H = 0.5$ mT.

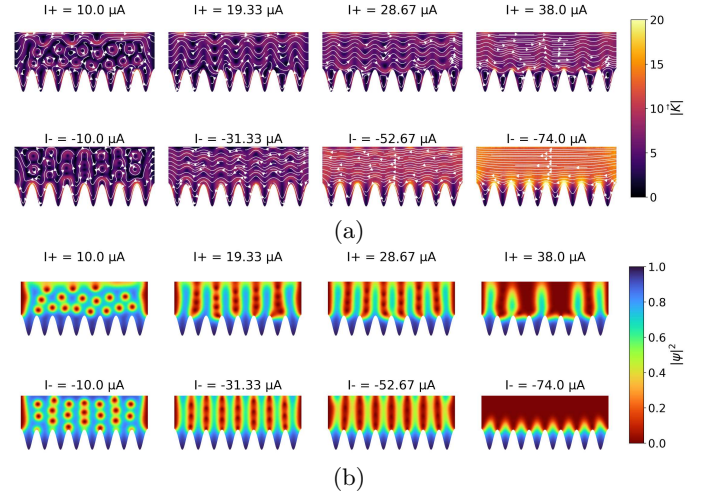


FIG. 14. Spatial distribution of (a) the supercurrent density and (b) the superconducting order parameter for Device 2 ($\kappa = 3$) exposed to a static magnetic field of $H = 0.5$ mT.

APPENDIX B. CRITICAL CURRENTS AND EFFICIENCY DATA

TABLE I. Critical current values and SDE rectification efficiency (η) for the four evaluated devices with a Ginzburg-Landau parameter $\kappa = 2$, as a function of the applied external magnetic field (H).

k	Device	H (mT)	J_{c1}^+/J_0	J_{c1}^-/J_0	J_{c2}^+/J_0	J_{c2}^-/J_0	J_c^+/J_0	J_c^-/J_0	η (%)
15*2	5*1	0,1	0,078	-0,067	0,155	-0,151	0,136	-0,132	1,42
		0,2	0,055	-0,044	0,147	-0,155	0,124	-0,147	8,45
		0,3	0,040	-0,032	0,139	-0,155	0,109	-0,147	14,92
		0,4	0,032	-0,021	0,128	-0,153	0,094	-0,145	21,60
		0,5	0,017	-0,017	0,097	-0,195	0,074	-0,145	32,17
	5*2	0,1	0,078	-0,082	0,151	-0,170	0,113	-0,094	9,25
		0,2	0,059	-0,044	0,151	-0,149	0,105	-0,143	15,38
		0,3	0,036	-0,032	0,143	-0,149	0,097	-0,143	19,04
		0,4	0,021	-0,017	0,128	-0,149	0,094	-0,143	20,96
		0,5	0,015	-0,008	0,111	-0,130	0,095	-0,122	12,28
	5*3	0,1	0,086	-0,090	0,158	-0,166	0,094	-0,097	2
		0,2	0,063	-0,074	0,151	-0,166	0,105	-0,120	6,77
		0,3	0,052	-0,063	0,143	-0,158	0,105	-0,078	14,63
		0,4	0,044	-0,059	0,132	-0,139	0,090	-0,067	14,63
		0,5	0,040	-0,052	0,097	-0,124	0,082	-0,071	7,5
	5*4	0,1	0,082	-0,086	0,151	-0,166	0,109	-0,097	5,55
		0,2	0,059	-0,071	0,151	-0,149	0,116	-0,143	10,29
		0,3	0,052	-0,059	0,143	-0,149	0,109	-0,143	13,63
		0,4	0,036	-0,044	0,109	-0,136	0,097	-0,128	13,55
		0,5	0,029	0,029	0,101	-0,120	0,094	-0,113	9,25

TABLE II. Critical current values and SDE rectification efficiency (η) for the four evaluated devices with a Ginzburg-Landau parameter $\kappa = 3$, as a function of the applied external magnetic field (H).

k	Device	H (mT)	J_{c1}^+/J_0	J_{c1}^-/J_0	J_{c2}^+/J_0	J_{c2}^-/J_0	J_c^+/J_0	J_c^-/J_0	η (%)
15*3	5*1	0,1	0,073	-0,082	0,159	-0,185	0,133	-0,159	8,82
		0,2	0,056	-0,064	0,150	-0,202	0,116	-0,176	20,59
		0,3	0,039	-0,047	0,142	-0,202	0,107	-0,176	24,24
		0,4	0,030	-0,039	0,133	-0,193	0,099	-0,176	28,12
		0,5	0,021	-0,021	0,125	-0,185	0,082	-0,159	32,14
	5*2	0,1	0,073	-0,073	0,159	-0,176	0,107	-0,159	19,35
		0,2	0,056	-0,039	0,159	-0,245	0,107	-0,176	24,24
		0,3	0,039	-0,039	0,150	-0,185	0,099	-0,168	25,81
		0,4	0,021	-0,030	0,133	-0,185	0,090	-0,159	27,58
		0,5	0,013	-0,013	0,125	-0,176	0,090	-0,159	27,58
	5*3	0,1	0,073	-0,082	0,168	-0,176	0,099	-0,116	8
		0,2	0,056	-0,064	0,159	-0,176	0,107	0,116	3,84
		0,3	0,039	-0,056	0,150	-0,168	0,099	-0,082	9,5
		0,4	0,030	-0,039	0,142	-0,142	0,090	-0,116	12,5
		0,5	0,026	-0,026	0,129	-0,137	0,086	-0,069	11,11
	5*4	0,1	0,073	-0,082	0,168	-0,168	0,107	-0,125	7,40
		0,2	0,047	-0,064	0,159	-0,168	0,116	-0,116	0,00
		0,3	0,039	-0,047	0,150	-0,185	0,099	-0,168	25,00
		0,4	0,030	-0,039	0,133	-0,142	0,099	-0,099	0,00
		0,5	0,021	-0,013	0,116	-0,090	0,099	-0,073	15,00

TABLE III. Critical current values and SDE rectification efficiency (η) for the four evaluated devices with a Ginzburg-Landau parameter $\kappa = 4$, as a function of the applied external magnetic field (H).

k	Device	H (mT)	J_{c1}^+/J_0	J_{c1}^-/J_0	J_{c2}^+/J_0	J_{c2}^-/J_0	J_c^+/J_0	J_c^-/J_0	η (%)
15*4	5*1	0,1	0,001	-0,053	0,176	-0,206	0,130	-0,160	10,53
		0,2	0,001	-0,038	0,160	-0,221	0,115	-0,176	21,05
		0,3	0,038	-0,023	0,160	-0,206	0,099	-0,176	27,78
		0,4	0,023	-0,008	0,145	-0,206	0,099	-0,160	23,53
		0,5	0,023	-0,008	0,130	-0,191	0,084	-0,160	31,25
	5*2	0,1	0,069	-0,069	0,176	-0,191	0,099	-0,160	23,53
		0,2	0,038	-0,053	0,160	-0,191	0,115	-0,160	16,67
		0,3	0,023	-0,038	0,160	-0,191	0,099	-0,160	23,53
		0,4	0,023	-0,023	0,145	-0,191	0,084	-0,160	31,25
		0,5	0,008	-0,008	0,130	-0,191	0,084	-0,160	31,25
	5*3	0,1	0,069	-0,069	0,176	-0,176	0,099	-0,130	13,33
		0,2	0,038	-0,053	0,176	-0,176	0,115	-0,115	0
		0,3	0,038	-0,038	0,160	-0,160	0,099	-0,084	8,33
		0,4	0,023	-0,023	0,145	-0,145	0,084	-0,115	15,38
		0,5	0,008	-0,023	0,130	-0,145	0,084	-0,115	15,38
	5*4	0,1	0,069	-0,069	0,176	-0,176	0,099	-0,115	7,14
		0,2	0,053	-0,053	0,176	-0,176	0,115	-0,115	0,00
		0,3	0,038	-0,038	0,160	-0,176	0,099	-0,084	8,30
		0,4	0,023	-0,038	0,145	-0,145	0,099	-0,099	0,00
		0,5	0,008	-0,023	0,130	-0,130	0,099	-0,099	0,00

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